Project Report

SOLAR ROOFTOP PV Plant Installation

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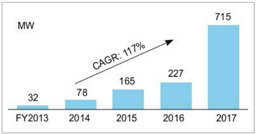
A guide to starting rooftop solar installation business in India

The Indian government has set its sights on having 175GW of non-hydro renewable capacity (made up of 60GW onshore wind, 60GW utility-scale solar, 10GW bio energy, 5GW small hydro and 40GW rooftop solar) by 2022. At present, this figure stands at just 60GW.

India is accelerating development of renewable energy projects to provide cheap, reliable and clean power to its 1.3 billion people. The country's per-capita on-grid electricity consumption increased by 22 per cent over the four years ending March 2017 due to increased industrial activity, higher uptake of electrical appliances by residential electricity users and addition of new consumers to the grid—according to a report by Bloomberg New Energy Finance (BNEF).

Majority of the capacity in the 2022 target—a total of 135GW—is utility-scale (wind farms and solar parks, in particular). The BNEF report estimates that India will need to invest $83 billion to build this capacity. The good news is that, because of falling capital costs per MW, that figure is $19 billion less than BNEF's previous estimates.

Fig. 1: Annual rooftop PV installations in India; India's financial year is from April to March (Source: Bloomberg New Energy Finance, industry surveys, Ministry of New and Renewable Energy)



But some of the most interesting activity will be in small-scale solar, that is, rooftop solar installation. According to BNEF, rooftop solar in India will grow inevitably with or without the support of power distribution utilities. India will need to invest $23 billion over the next five years to meet its 40GW rooftop photovoltaics target by year 2022.

The cost of electricity from rooftop PV has halved in the last five years due to fierce competition in the market and drop in equipment prices. In contrast, average retail electricity rates have increased by 22 per cent during the same period. This has made rooftop PV cheaper than commercial and industrial grid tariffs in all major states of India.

However, BNEF expects residential sector PV growth to pick up rapidly post-2021. At the moment, its attractiveness is being held back by the requirement of high upfront capital expenditure, shortage of financing options, and the fact that grid electricity is cheaper for residential consumers with low consumption.

BNEF experts also indicate that net metering is a far more important enabler for residential small-scale solar than business-scale projects.

Homeowners usually draw less power during the hours when their PV panels are producing, making self-consumption much harder. Solar projects also have a big and untapped potential to power irrigation pumps and reduce the use of back-up diesel